PAM-VLM: Disagreement-Aware Audio–Visual Emotion Recognition with Language-Model Reconciliation

Tuan-Anh Pham
Faculty of Electrical and Electronic Engineering
Phenikaa University
Hanoi, Vietnam
ptattmpxd01012004@gmail.com

***Abstract*—Audio–visual emotion recognition combines vocal and facial evidence, but published CREMA-D results are difficult to compare because the dataset has no official evaluation partition. This paper presents PAM-VLM, a multimodal architecture that uses pretrained WavLM and EfficientNet encoders, temporal visual modeling, bidirectional cross-attention, gated pooling, and a disagreement-aware Qwen reconciler. The primary experiment pools all 7,442 CREMA-D clips and constructs a deterministic, emotion-stratified 80/10/10 split containing 5,954 training, 744 validation, and 744 test samples. Clips are disjoint, while actors intentionally overlap across partitions. This protocol therefore measures within-corpus clip recognition rather than unseen-actor generalization. On the validation partition, selective fine-tuning obtains 90.61% Macro-F1. After model selection, one evaluation on the 744-clip test partition yields 88.82% Macro-F1, 88.56% unweighted average recall, and 88.71% accuracy. This improves test Macro-F1 by 3.93 points over the frozen hard-label fusion baseline and by more than 16 points over either unimodal model. In a separate 780/120-clip Mini-CREMA adaptation experiment, the reconciler raises validation Macro-F1 from 87.50% to 89.24%. On the held-out 744-clip CREMA-D test, however, routing lowers Macro-F1 from 88.70% to 88.17%. Zero-shot evaluation on all 1,056 prepared RAVDESS clips further decreases Macro-F1 from 56.18% for PAM to 50.71% with routing, exposing substantial cross-corpus domain shift.**

***Index Terms*—audio–visual emotion recognition, multimodal learning, cross-attention, CREMA-D, RAVDESS, large language model, label distribution learning**

I. Introduction

Emotion recognition supports affect-aware human–computer interaction and conversational systems. Speech contains prosody and voice quality, while video reveals facial configuration and temporal expression. Either modality may be ambiguous or degraded, so fusion must learn complementary evidence and stream reliability instead of merely concatenating two feature vectors.

CREMA-D [1] is widely used for this task. It contains 7,442 synchronized audio–visual clips from 91 actors expressing anger, disgust, fear, happiness, neutral affect, and sadness. Each utterance was rated under voice-only, face-only, and audio–visual presentation conditions. This annotation design supports both conventional hard-label classification and learning from perceptual disagreement.

CREMA-D has no official split. Published studies use random clip partitions, speaker-independent folds, different trial counts, and different metrics. A clip-level score can benefit from actors shared with training, whereas an actor-independent score measures transfer to unseen identities. The protocols answer different questions and are not directly interchangeable.

This work targets the highest measured recognition quality under a fixed clip-level protocol. All clips are pooled and divided with a deterministic, emotion-stratified 80/10/10 split. A clip appears in exactly one partition, but an actor may occur in training, validation, and test. The resulting benchmark uses more training data than an actor-disjoint setting and is aligned with the random clip-level family used by several prior studies [2]–[4]. The limitation is explicit: the score does not estimate performance on unseen speakers.

PAM-VLM combines WavLM speech representations [5] and EfficientNet facial representations [6] in a shared 256-dimensional space. A temporal Transformer models the sampled face sequence. Bidirectional cross-attention transfers context in both directions, and a feature-wise gate produces the fused embedding. Separate audio, video, and fusion heads enable controlled unimodal comparisons and modality-matched supervision. A secondary language-model branch maps fused and branch-specific evidence into soft prefix tokens for constrained Qwen reconciliation.

The contributions of this study are:

- a reproducible 80/10/10 clip-level CREMA-D protocol with exact class counts, explicit actor overlap, and disjoint clips;
- a cross-attention architecture that selectively fine-tunes upper pretrained layers while retaining a compact 18.08M-parameter trainable subset;
- a disagreement-aware Qwen reconciler with constrained six-label scoring and a validation-selected confidence guard;
- a complete experimental setup covering media processing, augmentation, optimization, checkpoint selection, and evaluation metrics; and
- a final test result of 88.82% Macro-F1, with controlled unimodal and frozen-fusion comparisons under the same

split.

II. Related Work and Evaluation Context

A. Audio–Visual Fusion

Transformer fusion learns dependencies between speech and facial behavior. Ghaleb *et al.* introduced metric-learning fusion [7] and later combined temporal encoding with modality attention in MATER [8]. Birhala *et al.* [9] aggregate asynchronous audio and visual dynamics. Tran and Soleymani [10] pretrain an audio–visual Transformer on VoxCeleb2, while AuxFormer [11] adds auxiliary unimodal objectives so that one branch does not erase the information learned by the other. VAVL [12] combines shared conformer layers and reconstruction losses.

Recent work explores several complementary directions. OTM [2] regulates modality contributions during training; Mocanu *et al.* [3] combine cross-modal attention and metric learning; and HiCMAE [13] applies hierarchical contrastive masked pretraining. DE-III [14] enhances both intra- and inter-modal detail, and Wang *et al.* [15] use multi-head cross-attention with staged training. Lightweight uncertainty learning [16], shared cross-attention Transformers [17], supervised feature mapping [18], and unified intra/inter-modal interaction [4] further broaden the CREMA-D evidence base. PAM focuses on selective adaptation of established encoders with a compact bidirectional fusion stack.

B. Perceptual Label Distributions

A single categorical target hides disagreement among annotators. Label distribution learning [19] instead represents the degree to which each label describes a sample. CREMA-D is particularly suitable because its ratings were collected under three presentation conditions. Goncalves *et al.* [20] show that jointly learning from unimodal-rated and multimodal-rated labels provides complementary supervision. PAM therefore associates the voice distribution with the audio head, the face distribution with the video head, and the audio–visual distribution with the fusion head.

C. Protocol Comparability

Table I gives a compact CREMA-D comparison. The selected studies are peer-reviewed and cover both identity-independent and clip-level evaluation. Only the latter group is numerically close to the primary protocol, although none uses PAM's exact manifest.

III. Method

A. Overview

Figure 1 shows the complete design. The benchmark path contains an audio encoder, a video encoder, bidirectional fusion, and three classification heads. The language-model path converts the fused representation and branch predictions into soft prefix tokens for disagreement-aware reconciliation. The 744-clip CREMA-D benchmark result uses the PAM classifier, while the reconciler is evaluated separately on the Mini-CREMA adaptation split, the full CREMA-D test partition, and the complete prepared RAVDESS zero-shot set.

B. Audio and Video Encoders

Let the waveform input be $\mathbf{x}^a \in \mathbb{R}^{B\times T}$. WavLM-base-plus produces a sequence of 768-dimensional speech states. A linear layer, layer normalization, and dropout project these states to $\mathbf{A} \in \mathbb{R}^{B\times T_a\times d}$ with $d = 256$. Padding masks are propagated through WavLM and used by every subsequent attention and pooling operation.

For video, 16 RGB frames are sampled from each clip. Each detected face is resized to 224×224 and encoded independently by EfficientNet-B0, producing a 1280-dimensional frame representation. A linear projection maps the frame features to $d = 256$. A two-layer Transformer encoder with four attention heads and GELU feed-forward blocks models facial motion across the sampled sequence:

$$\mathbf{V} = \mathrm{Transformer}_2\left(\mathrm{Proj}_v(\mathrm{EffNet}(\mathbf{x}^v))\right). \tag{1}$$

The selected model freezes the lower pretrained blocks and fine-tunes only the final two WavLM Transformer layers and final two EfficientNet feature stages. Projection, temporal, fusion, normalization, and classification modules remain fully trainable. The model contains 101.16M parameters, of which 18.08M are updated.

C. Bidirectional Cross-Attention and Gating

Audio and video sequences have different lengths and are not assumed to be index-aligned. Two multi-head attention modules exchange context:

$$\mathbf{A}' = \mathrm{LN}\left(\mathbf{A} + \mathrm{MHA}(\mathbf{A}, \mathbf{V}, \mathbf{V})\right), \tag{2}$$

$$\mathbf{V}' = \mathrm{LN}\left(\mathbf{V} + \mathrm{MHA}(\mathbf{V}, \mathbf{A}, \mathbf{A})\right). \tag{3}$$

The first expression asks which facial states support each speech state; the second asks which speech states support each facial state. Invalid positions are masked before attention. Masked temporal means then produce $\bar{\mathbf{a}}$ and $\bar{\mathbf{v}}$.

A feature-wise gate determines the mixture:

$$\mathbf{g} = \sigma(\mathbf{W}_g[\bar{\mathbf{a}}; \bar{\mathbf{v}}] + \mathbf{b}_g), \quad \mathbf{e} = \mathrm{LN}\left(\mathbf{g} \odot \bar{\mathbf{a}} + (1 - \mathbf{g}) \odot \bar{\mathbf{v}}\right). \tag{4}$$

Three dropout-linear heads map $\bar{\mathbf{a}}$, $\bar{\mathbf{v}}$, and $\mathbf{e}$ to six-class audio, video, and fusion logits. The fusion head is the benchmark output.

D. Joint Hard and Perceptual Objective

The selected model combines label-smoothed categorical supervision with modality-matched perceptual distributions. Its hard-label component is

$$\mathcal{L}_{hard} = 0.75\mathcal{L}_{CE}^{av} + 0.2\mathcal{L}_{CE}^{a} + 0.05\mathcal{L}_{CE}^{v}, \tag{5}$$

where cross-entropy uses label smoothing 0.05. The larger fusion weight prioritizes the deployed output while retaining branch-specific supervision.

For the perceptual extension, vote counts $n_c^{(m)}$ under condition m are converted to Laplace-smoothed distributions:

$$\tilde{p}_c^{(m)} = \frac{n_c^{(m)} + \alpha}{\sum_{j=1}^{6} n_j^{(m)} + 6\alpha}, \qquad \alpha = 1. \tag{6}$$

Table I
COMPACT COMPARISON WITH PEER-REVIEWED AUDIO–VISUAL CREMA-D STUDIES. SCORES RETAIN EACH PAPER'S PROTOCOL AND METRIC. SI DENOTES SPEAKER-INDEPENDENT EVALUATION.

Study	Protocol	Metric	Score (%)
Identity-independent evaluation			
Tran and Soleymani [10]	SI 60/20/20	Accuracy	70.22
AuxFormer [11]	SI 70/15/15, 20 trials	Macro-F1	71.30
VAVL [12]	SI 70/15/15, five trials	Macro-F1	77.90 ± 2.50
HiCMAE [13]	SI five-fold	UAR/WAR	84.91/84.89
Clip-level or source-defined evaluation			
OTM [2]	Random 6,698/744	Accuracy	75.10
Wang *et al.* [15]	Random 70/15/15, five trials	Macro-F1/Accuracy	84.50/88.29
FMII [4]	Random 80/20	Accuracy	88.21 ± 0.34
PAM (ours)	**Clip 80/10/10, test**	**Macro-F1/Accuracy**	**88.82/88.71**

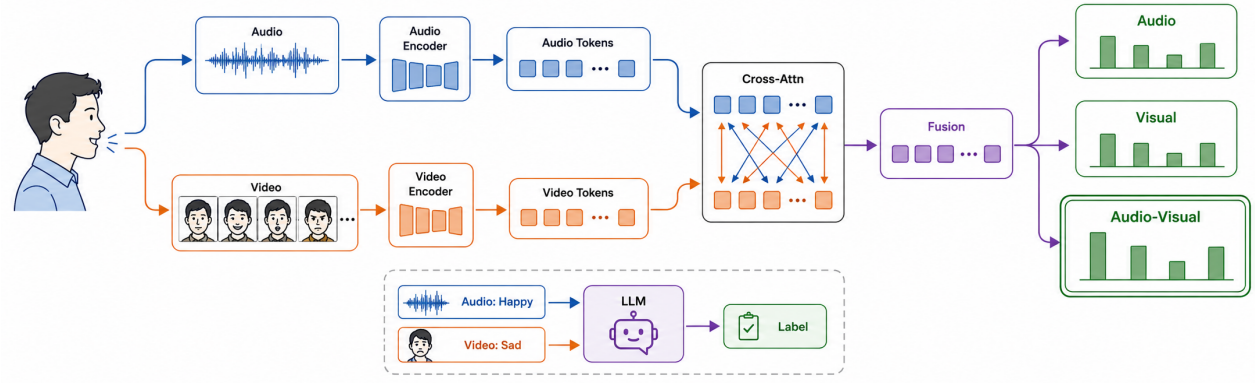


Figure 1. PAM processing pipeline. WavLM audio tokens and temporally encoded EfficientNet face tokens interact through bidirectional cross-attention. Audio, video, and fused heads predict six emotions. The dashed projector and Qwen path reconciles these predictions when the unimodal branches disagree.

If a clip-condition pair has no votes, its intended hard emotion is used as an explicit fallback. The final objective is

$$\mathcal{L} = \mathcal{L}_{hard} + 0.15 D_{KL}(\tilde{p}_{av} \| p_{av}) + 0.05 D_{KL}(\tilde{p}_{voice} \| p_a) + 0.02 D_{KL}(\tilde{p}_{face} \| p_v). \quad (7)$$

These terms preserve a sharp categorical anchor while transferring human perceptual disagreement to the matching branch.

E. Language-Model Reconciler and Safety Routing

The dashed branch in Fig. 1 is designed for interpretable reconciliation rather than replacing the classifier. A projector concatenates the 256-dimensional fusion embedding, the two six-class unimodal probability vectors, and one audio–video disagreement scalar. Two linear layers reshape the resulting 269-dimensional vector into four prefix embeddings. These are prepended to the prompt embeddings of Qwen2.5-0.5B-Instruct [21]. The base language model is frozen and rank-16 LoRA adapters [22] are attached to its query, key, value, and output projections.

The textual prompt names the six-class audio, video, and fusion probability vectors together with the disagreement value. During adaptation, common prompt tokens are masked and supervision is applied only to the target emotion tag. At inference, each of the six valid tags is scored by conditional negative log-likelihood, preventing unconstrained generations from changing the label space. The four 896-dimensional prefix tokens and 2.16M LoRA parameters are trainable; the PAM encoders and Qwen base weights remain frozen.

Because a reconciler can degrade reliable classifier decisions, PAM-VLM uses a validation-selected safety rule. Qwen is consulted only when audio–video disagreement is at least 0.324 and PAM confidence is at most 0.665; otherwise the fusion prediction is retained. The rule is selected on Mini-CREMA validation data and transferred unchanged to RAVDESS.

IV. EXPERIMENTAL SETUP

This section states the complete setup used to produce the reported values. The distinction between training, model selection, and final evaluation is maintained throughout.

A. Dataset and Clip-Level Split

The input corpus is the complete CREMA-D release: 7,442 clips, 91 actors, six emotion classes, and paired audio/video media. The primary split is generated once with seed 42 and stratification by hard emotion. The five non-neutral classes

each contain 1,271 clips; neutral contains 1,087. Stratification produces 5,954 training, 744 validation, and 744 test clips.

No clip ID is shared across partitions. All 91 actors occur in all three partitions by design, so identity overlap is not accidental leakage but a declared property of the protocol. The split answers: "How accurately can the model classify unseen clips drawn from the same actor population?" It does not answer: "How accurately does the model generalize to a new actor?"

The split is created once before training and then held fixed for every model. Class identities and clip membership are unchanged across the audio-only, video-only, and fusion experiments. Test labels are never used for tuning or checkpoint selection.

B. Media Preprocessing

Audio is decoded as a mono waveform and resampled to 16 kHz. Variable-length waveforms are padded within each mini-batch, and a Boolean mask prevents padded samples from contributing to WavLM attention or temporal pooling. No spectrogram is computed because WavLM consumes waveform samples directly.

Video is decoded with OpenCV. During training, 16 frame indices are sampled without replacement when enough frames are available; validation uses 16 deterministic linearly spaced indices. An offline Haar cascade detects a face in each frame. The detected square is expanded by a 15% margin and resized to 224×224. A center-square crop is used when detection fails. RGB values are converted to floating point in $[0, 1]$.

Augmentation is restricted to the training partition and applied with probability 0.8 independently for each modality pipeline. Audio augmentation uses a circular shift of at most 10% of waveform length, gain in $[0.9, 1.1]$, Gaussian noise with standard deviation 0.005 times signal RMS, and polarity inversion with probability 0.5. Video augmentation applies one clip-consistent horizontal flip with probability 0.5 and brightness/contrast jitter of at most 0.2. Validation and test media receive no random augmentation.

C. Model and Optimization Settings

Table II consolidates the model and training setup. All reported models use the same split, preprocessing, augmentation policy, and seed. The audio-only, video-only, and frozen-fusion runs provide controlled baselines. The selected high-accuracy model then changes the fine-tuning and regularization settings listed below; its contribution is therefore measured relative to the frozen-fusion baseline as well as the unimodal models.

AdamW uses 2×10^{-4} for task and fusion layers and 10^{-5} for the unfrozen backbone blocks. Cosine annealing reduces the task learning rate to 10^{-6}. A physical batch size of 4 with eight accumulation steps gives an effective batch size of 32. Mixed precision is enabled and the gradient norm is clipped at 1.0. The best validation Macro-F1 occurs at zero-indexed epoch 24. Training stops after epoch 32 when eight subsequent epochs fail to improve the selection metric.

Table II
IMPLEMENTATION AND TRAINING SETUP FOR THE REPORTED PRIMARY RUNS.

Component	Configuration
Audio input	16-kHz waveform; masked variable-length batches
Video input	16 RGB face frames, 224×224; 15% crop margin
Audio encoder	WavLM-base-plus; final two layers trainable; 768-D to 256-D
Video encoder	EfficientNet-B0; final two stages trainable; 1280-D to 256-D
Fusion	Two 4-head cross-attention directions; audio gate bias 1.5
Regularization	Fusion dropout 0.15; head dropout 0.25; modality dropout 0.2
Objective	Weighted CE with smoothing 0.05 and modality-matched KL
Optimization	AdamW; task LR 2×10^{-4}; backbone LR 10^{-5}; WD 0.01
Training budget	At most 40 epochs; batch 4; accumulation 8; patience 8
Model selection	Maximum validation Macro-F1; epoch 24 selected

D. Reconciler Adaptation and Transfer Protocol

The language branch is adapted on a balanced Mini-CREMA subset containing 780 training clips (130 per class) and 120 validation clips (20 per class). This subset is used only to establish whether the projector, LoRA adapters, and safety routing improve a fixed PAM classifier; it contains no samples from the primary 744-clip CREMA-D test partition. PAM features and probability vectors are cached, while Qwen LoRA parameters and the projector are optimized with learning rates of 2×10^{-5} and 2×10^{-4}, respectively. The physical batch size is 4 with four accumulation steps. Epoch 1 is selected by validation Macro-F1.

Cross-corpus transfer uses RAVDESS [23], restricted to the six labels shared with CREMA-D. The complete prepared evaluation set contains 1,056 clips from 24 actors. No RAVDESS labels are used for adaptation or threshold selection. The frozen PAM, Qwen reconciler, and safety router are evaluated once on all 1,056 clips. The same Mini-CREMA-selected routing thresholds are also evaluated on the untouched 744-clip CREMA-D test set.

E. Evaluation Metrics and Decision Rule

For class c, precision and recall are

$$P_c = \frac{TP_c}{TP_c + FP_c}, \qquad R_c = \frac{TP_c}{TP_c + FN_c}. \tag{8}$$

Class F1 is $F1_c = 2P_cR_c/(P_c + R_c)$. With $C = 6$, the two primary metrics are

$$\text{MacroF1} = \frac{1}{C}\sum_{c=1}^{C} F1_c, \qquad \text{UAR} = \frac{1}{C}\sum_{c=1}^{C} R_c. \tag{9}$$

Accuracy is the number of correct predictions divided by the number of samples. Macro-F1 is the checkpoint-selection metric because it weights all classes equally and penalizes both low precision and low recall. UAR isolates recall balance, while accuracy preserves comparability with papers that report

only recognition rate. Calibration is summarized with 15-bin expected calibration error (ECE):

$$\mathrm{ECE} = \sum_{b=1}^{15} \frac{|B_b|}{N} \left|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)\right|. \tag{10}$$

The development decision rule is fixed before testing: compare candidate models on validation Macro-F1, freeze the selected architecture and hyperparameters, and evaluate the 744-clip test partition once. This procedure has now been completed; the primary numbers below are held-out test results.

V. Results

A. Training and Model Selection

Figure 2 summarizes optimization of the selected model. The training loss decreases throughout the run, while validation loss falls rapidly and then stabilizes near 0.72. Validation Macro-F1 increases from approximately 0.73 to 0.906. At the selected epoch 24, train and validation Macro-F1 are 0.987 and 0.906, respectively, and train/validation losses are 0.570/0.725. The gap indicates residual overfitting, but validation performance does not collapse. Early stopping after epoch 32 prevents further optimization after eight non-improving epochs.

B. Overall Validation and Test Results

The final test results in Table III confirm the validation-based selection. PAM reaches 0.888 Macro-F1, 0.886 UAR, and 0.887 accuracy. Relative to frozen hard-label fusion, selective fine-tuning and the joint objective add 3.93 Macro-F1, 4.02 UAR, and 4.03 accuracy points. The gain over audio-only is 16.78 Macro-F1 points, and the gain over video-only is 18.59 points. Validation Macro-F1 exceeds test Macro-F1 by 1.80 points, a moderate selection gap rather than a reversal of model ranking.

The improvement has a calibration cost. ECE increases from 0.038 for frozen fusion to 0.090 for the high-accuracy model, even though its classification metrics are substantially better. Temperature scaling on a separate calibration subset is therefore a suitable deployment step, but it must not be fit on the held-out test set.

C. Language Reconciliation and Cross-Corpus Transfer

On Mini-CREMA, direct Qwen reconciliation improves Macro-F1 by 1.69 points over PAM, and safety routing reaches 0.8924. This is validation evidence rather than an independent test result because the checkpoint and thresholds are selected on the same 120 samples.

The held-out evaluations reverse that ordering. On CREMA-D test, direct Qwen reduces Macro-F1 by 1.20 points and routing reduces it by 0.53 points relative to the cached PAM baseline while routing 60.6% of clips. On RAVDESS, PAM obtains 0.5618 Macro-F1, while Qwen and routing obtain 0.5004 and 0.5071, respectively. Routing is activated for 35.4% of RAVDESS clips. The result shows that Mini-CREMA confidence and disagreement thresholds do not generalize reliably to held-out or cross-corpus samples.

Table V places the zero-shot result near established audio–visual methods without claiming direct superiority. The published studies train and test on RAVDESS, whereas PAM receives no RAVDESS adaptation and uses only the six labels shared with CREMA-D. The substantially lower score therefore measures cross-corpus transfer rather than within-RAVDESS supervised recognition.

D. Class-Level Analysis

Happiness is the most reliable class, with 0.972 F1 and 121/127 correct predictions. Angry and disgust both reach approximately 0.918 F1. Neutral has high precision (0.957) but lower recall (0.826), whereas sadness has the lowest precision (0.750) and 0.874 recall. The close aggregate values for Macro-F1, UAR, and accuracy indicate that the final score is not produced by one dominant class.

Figure 3 shows that the principal residual error is attraction toward sadness: 17 fear and 11 neutral clips are predicted as sad. In the opposite direction, nine sad clips are predicted as fear. This pattern explains sadness's low precision and fear's lower recall. Class-conditional reliability modeling should therefore focus on the fear–sad and neutral–sad boundaries.

E. Position Relative to Published Work

Within the random or clip-level family, PAM's 88.71% test accuracy is numerically above OTM's 75.10%, Wang *et al.*'s 88.29%, and FMII's $88.21 \pm 0.34\%$. PAM's 88.82% Macro-F1 is also above the 84.50% reported by Wang *et al.* These values define context rather than universal superiority because even the closest studies differ in split ratios, seed averaging, feature encoders, and model-selection procedures.

Speaker-independent and cross-validation results in the first block of Table I answer a harder identity-generalization question and are deliberately not ranked against PAM. A strict comparison requires retraining accessible baselines on the PAM manifest over matched seeds. The present evidence supports strong clip-level within-corpus recognition, not a protocol-independent state-of-the-art claim.

VI. Discussion and Future Work

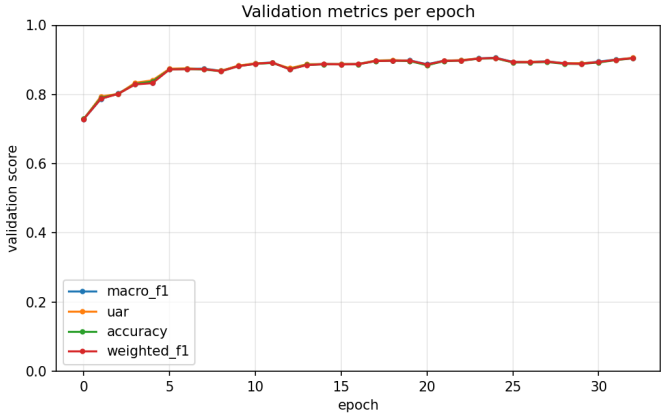
A. Effect of Selective Fine-Tuning

The 3.93-point test Macro-F1 gain over frozen fusion shows that generic pretrained representations remain improvable for acted emotion recognition. Restricting updates to upper backbone blocks adapts high-level acoustic and facial cues without optimizing all 101.16M parameters. Modality dropout and the audio-biased gate further discourage dependence on one stream, while the KL terms connect each branch to the matching perceptual condition.

The experiment changes several components together, so the gain cannot yet be attributed to one mechanism. A factorial ablation should add selective fine-tuning, modality dropout, gate bias, label smoothing, and perceptual KL sequentially



(a) Training and validation objective.



(b) Validation classification metrics.

Figure 2. Learning curves for the selected high-accuracy model. The best validation Macro-F1 occurs at epoch 24; training terminates after epoch 32 under patience 8.

Table III
VALIDATION AND HELD-OUT TEST RESULTS UNDER THE SHARED CLIP-LEVEL 80/10/10 SPLIT. ECE USES 15 CONFIDENCE BINS. BOLD DENOTES THE BEST TEST VALUE.

Model	Val. Macro-F1	Test Macro-F1	Test UAR	Test Accuracy	Test ECE
Audio-only	0.738	0.720	0.720	0.719	0.057
Video-only	0.736	0.702	0.697	0.696	0.062
Frozen fusion, hard labels	0.832	0.849	0.845	0.847	**0.038**
PAM high-accuracy	**0.906**	**0.888**	**0.886**	**0.887**	0.090

Table IV
PAM-VLM RECONCILIATION RESULTS. MINI-CREMA IS VALIDATION DATA USED FOR CHECKPOINT AND ROUTING SELECTION; CREMA-D TEST AND RAVDESS ZERO-SHOT ARE FULL EVALUATIONS WITH FROZEN THRESHOLDS.

Evaluation	Decision system	Samples	Macro-F1	UAR	Accuracy
Mini-CREMA validation	PAM	120	0.8750	0.8750	0.8750
	Qwen reconciler	120	0.8919	0.8917	0.8917
	PAM+VLM routing	120	**0.8924**	**0.8917**	**0.8917**
CREMA-D held-out test	PAM cache baseline	744	**0.8870**	**0.8846**	**0.8858**
	Qwen reconciler	744	0.8750	0.8727	0.8737
	PAM+VLM routing	744	0.8817	0.8791	0.8804
RAVDESS zero-shot	PAM	1,056	**0.5618**	**0.5573**	**0.5634**
	Qwen reconciler	1,056	0.5004	0.5226	0.5312
	PAM+VLM routing	1,056	0.5071	0.5278	0.5369

under the same split. This is more informative than treating the combined high-accuracy configuration as evidence that every component helps.

B. Error-Driven Improvements

The confusion pattern suggests tracked, aligned face crops to reduce temporal jitter and class-conditional reliability gates that can favor speech for one emotion and facial motion for another. A confusion-aware loss is justified only if repeated seeds confirm the sadness false-positive pattern.

Improvements should be staged: selective fine-tuning, modality dropout, perceptual KL supervision, then reliability gating. Each stage uses the same split and reports its incremental Macro-F1. Three to five seeds should estimate variance before a final ensemble is considered.

C. Role of the Language Model

The Mini-CREMA result shows that language-model reconciliation can fit a small subset of uncertain in-domain decisions. The two full evaluations show that this gain does not survive held-out testing or cross-corpus transfer. Routing more samples is harmful when confidence and disagreement distributions shift. Qwen is not a native audio–visual foundation model in this system. It receives compact PAM evidence through learned prefix tokens and returns one of six constrained labels.

Required ablations are Qwen without LoRA, LoRA without routing, alternative calibration scores, and routing learned on a larger held-out development set. Any new router must be selected without test labels. Generated rationales remain

Table V
PUBLISHED AUDIO–VISUAL RAVDESS CONTEXT. PROTOCOLS AND LABEL SETS DIFFER, SO THE VALUES ARE NOT A COMMON LEADERBOARD.

Study	**Evaluation protocol**	**Labels**	**Accuracy (%)**
MATER [8]	Paper-defined audio–visual evaluation	Paper-defined	76.30
Luna-Jimenez *et al.* [24]	Subject-wise five-fold	8 classes	80.08
Middya *et al.* [25]	Random 80/10/10	8 classes	86.00
Mocanu *et al.* [3]	Paper-defined audio–visual evaluation	Paper-defined	89.25
CAMS [17]	Paper-defined audio–visual evaluation	Paper-defined	89.78
PAM zero-shot	All 1,056 clips; no RAVDESS adaptation	6 classes	56.34
PAM+VLM zero-shot	All 1,056 clips; no RAVDESS adaptation	6 classes	53.69

Table VI
PER-CLASS PERFORMANCE OF PAM HIGH-ACCURACY ON THE 744-CLIP TEST SET.

Emotion	**Precision**	**Recall**	**F1**	**Support**
Angry	0.914	0.921	0.918	127
Disgust	0.908	0.929	0.918	127
Fear	0.844	0.811	0.827	127
Happy	0.992	0.953	0.972	127
Neutral	0.957	0.826	0.887	109
Sad	0.750	0.874	0.807	127

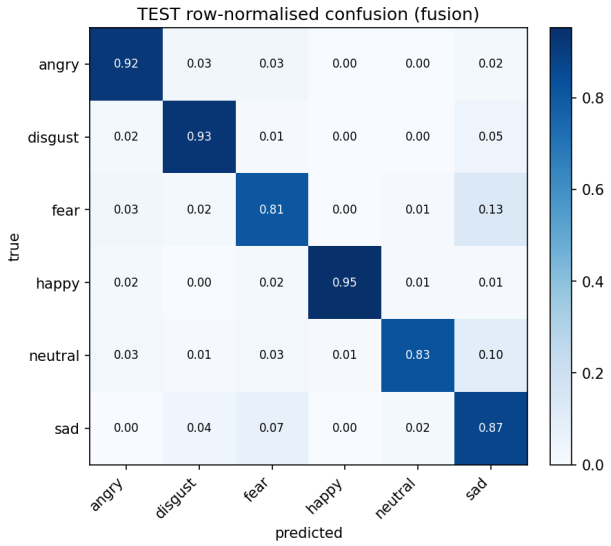


Figure 3. Row-normalized confusion matrix of PAM high-accuracy on the held-out test partition. Rows are true classes and columns are predictions.

secondary to label accuracy.

VII. LIMITATIONS AND THREATS TO VALIDITY

The main limitation is deliberate actor overlap. Recurring voice, face, or sentence cues may make the task easier than unseen-speaker evaluation. The result comes from one corpus and one split seed; a different partition may change it. Although the test partition was evaluated only after model selection, repeated seeds are still required to estimate variance.

CREMA-D contains acted English recorded under controlled conditions. Performance may not transfer to spontaneous emotion, other languages, noisy media, or asynchronous streams. Haar detection is reproducible but less robust than modern alignment, and missing-stream robustness remains unmeasured.

External validity is further limited by protocol mismatch across published work. Some papers average multiple speaker-independent folds, while others use one random clip split. Accuracy, UAR, and Macro-F1 are not interchangeable. For these reasons, Table I is descriptive and the report does not claim a universal ranking.

The current language-model evidence has two additional boundaries. Mini-CREMA reports the same validation set used for checkpoint and routing selection, so it is not a held-out performance claim. The RAVDESS evaluation covers the complete prepared set, but its six-class mapping and zero-shot CREMA-D training differ from the supervised eight-class protocols in prior work. Neither language-model result replaces the primary PAM test score.

VIII. ETHICAL CONSIDERATIONS

Emotion labels are perceptual judgments, not direct measurements of internal state. Predictions reflect acting, annotation, demographic, and recording biases. The system must not be the sole basis for high-stakes decisions. Deployment requires subgroup evaluation, consent, privacy protection, and a way to contest inferences. Generated rationales are not factual explanations of a person's mental state.

IX. CONCLUSION

PAM-VLM combines pretrained audio and visual encoders with temporal modeling, bidirectional cross-attention, and gated pooling for six-class CREMA-D emotion recognition. Under a deterministic, emotion-stratified 80/10/10 clip-level split, the selected high-accuracy model reaches 90.61% validation Macro-F1. On the held-out test partition it obtains 88.82% Macro-F1, 88.56% UAR, and 88.71% accuracy. This improves test Macro-F1 by 3.93 points over frozen hard-label fusion and by more than 16 points over both unimodal models. The setup, learning curves, class-level errors, and selection rule are reported so that the experiment can be reproduced and interpreted correctly.

On Mini-CREMA validation, disagreement-aware reconciliation raises Macro-F1 from 87.50% to 89.24%. This gain does not hold on the complete evaluations: routing reduces

CREMA-D test Macro-F1 from 88.70% to 88.17% and RAVDESS zero-shot Macro-F1 from 56.18% to 50.71%. The negative result identifies validation overfitting and unresolved cross-corpus shift in the reconciler.

The primary result demonstrates strong within-corpus clip recognition, not generalization to unseen actors. Multi-seed confirmation, component-wise ablation, larger routing-development data, cross-corpus calibration, and stable face tracking are the most direct next steps.

REFERENCES

[1] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "CREMA-D: Crowd-sourced emotional multimodal actors dataset," *IEEE Transactions on Affective Computing*, vol. 5, no. 4, pp. 377–390, 2014.

[2] Y. Wei, D. Hu, H. Du, and J.-R. Wen, "On-the-fly modulation for balanced multimodal learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 1, pp. 469–485, 2025.

[3] B. Mocanu, R. Tapu, and T. Zaharia, "Multimodal emotion recognition using cross modal audio-video fusion with attention and deep metric learning," *Image and Vision Computing*, vol. 133, p. 104676, 2023.

[4] Z. Sun, H. Liu, Y. Li, H. Li, W. Zhang, and T. Song, "FMII: A unified intra- and inter-modal interaction framework for robust audio-visual emotion recognition," *Journal of King Saud University Computer and Information Sciences*, vol. 38, p. 180, 2026.

[5] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, J. Wu, M. Zeng, X. Yu, and F. Wei, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.

[6] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proceedings of the 36th International Conference on Machine Learning*, 2019, pp. 6105–6114.

[7] E. Ghaleb, M. Popa, and S. Asteriadis, "Metric learning-based multimodal audio-visual emotion recognition," *IEEE MultiMedia*, vol. 27, no. 1, pp. 37–48, 2020.

[8] E. Ghaleb, J. Niehues, and S. Asteriadis, "Joint modelling of audio-visual cues using attention mechanisms for emotion recognition," *Multimedia Tools and Applications*, vol. 82, no. 8, pp. 11 239–11 264, 2023.

[9] A. Birhala, C. N. Ristea, A. Radoi, and L. C. Dutu, "Temporal aggregation of audio-visual modalities for emotion recognition," in *Proceedings of the 43rd International Conference on Telecommunications and Signal Processing*, 2020, pp. 305–308.

[10] M. Tran and M. Soleymani, "A pre-trained audio-visual transformer for emotion recognition," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 4698–4702.

[11] L. Goncalves and C. Busso, "AuxFormer: Robust approach to audio-visual emotion recognition," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 7357–7361.

[12] L. Goncalves, S.-G. Leem, W.-C. Lin, B. Sisman, and C. Busso, "Versatile audio-visual learning for emotion recognition," *IEEE Transactions on Affective Computing*, vol. 16, no. 1, pp. 306–318, 2025.

[13] L. Sun, Z. Lian, B. Liu, and J. Tao, "HiCMAE: Hierarchical contrastive masked autoencoder for self-supervised audio-visual emotion recognition," *Information Fusion*, vol. 108, p. 102382, 2024.

[14] T. Shi, X. Ge, J. M. Jose, N. Pugeault, and P. Henderson, "Detail-enhanced intra- and inter-modal interaction for audio-visual emotion recognition," in *Proceedings of the International Conference on Pattern Recognition*, ser. Lecture Notes in Computer Science, vol. 15321, 2024, pp. 451–465.

[15] Z. Wang, D. Zhao, L. Qin, Y. Chen, and Y. Shen, "Audiovisual emotion recognition based on a multi-head cross attention mechanism," *Journal of Biomedical Engineering*, vol. 42, no. 1, pp. 24–31, 2025.

[16] A. Radoi and G. Cioroiu, "Uncertainty-based learning of a lightweight model for multimodal emotion recognition," *IEEE Access*, vol. 12, pp. 120 362–120 374, 2024.

[17] J. Lei, K. Ye, and Y. Wang, "Cross-attention fusion for audio-visual emotion recognition with shared transformer," *Speech Communication*, vol. 178, p. 103356, 2026.

[18] M. H. Tayarani Najaran, S. I. Shahid, F. Foerster, and V. Steuber, "A deep learning feature mapping algorithm for emotion detection via facial and audio signals," *Applied Soft Computing*, vol. 197, p. 114998, 2026.

[19] X. Geng, "Label distribution learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 28, no. 7, pp. 1734–1748, 2016.

[20] L. Goncalves, H.-C. Chou, A. N. Salman, C.-C. Lee, and C. Busso, "Jointly learning from unimodal and multimodal-rated labels in audio-visual emotion recognition," *IEEE Open Journal of Signal Processing*, vol. 6, pp. 165–174, 2025.

[21] Qwen Team, "Qwen2.5: A party of foundation models," 2025, arXiv:2412.15115.

[22] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations*, 2022.

[23] S. R. Livingstone and F. A. Russo, "The ryerson audio-visual database of emotional speech and song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in north american english," *PLOS ONE*, vol. 13, no. 5, p. e0196391, 2018.

[24] C. Luna-Jimenez, D. Griol, Z. Callejas, R. Kleinlein, J. M. Montero, and F. Fernandez-Martinez, "Multimodal emotion recognition on RAVDESS dataset using transfer learning," *Sensors*, vol. 21, no. 22, p. 7665, 2021.

[25] A. I. Middya, B. Nag, and S. Roy, "Deep learning based multimodal emotion recognition using model-level fusion of audio–visual modalities," *Knowledge-Based Systems*, vol. 244, p. 108580, 2022.